

PROJECT DESIGN PHASE

Solution Architecture

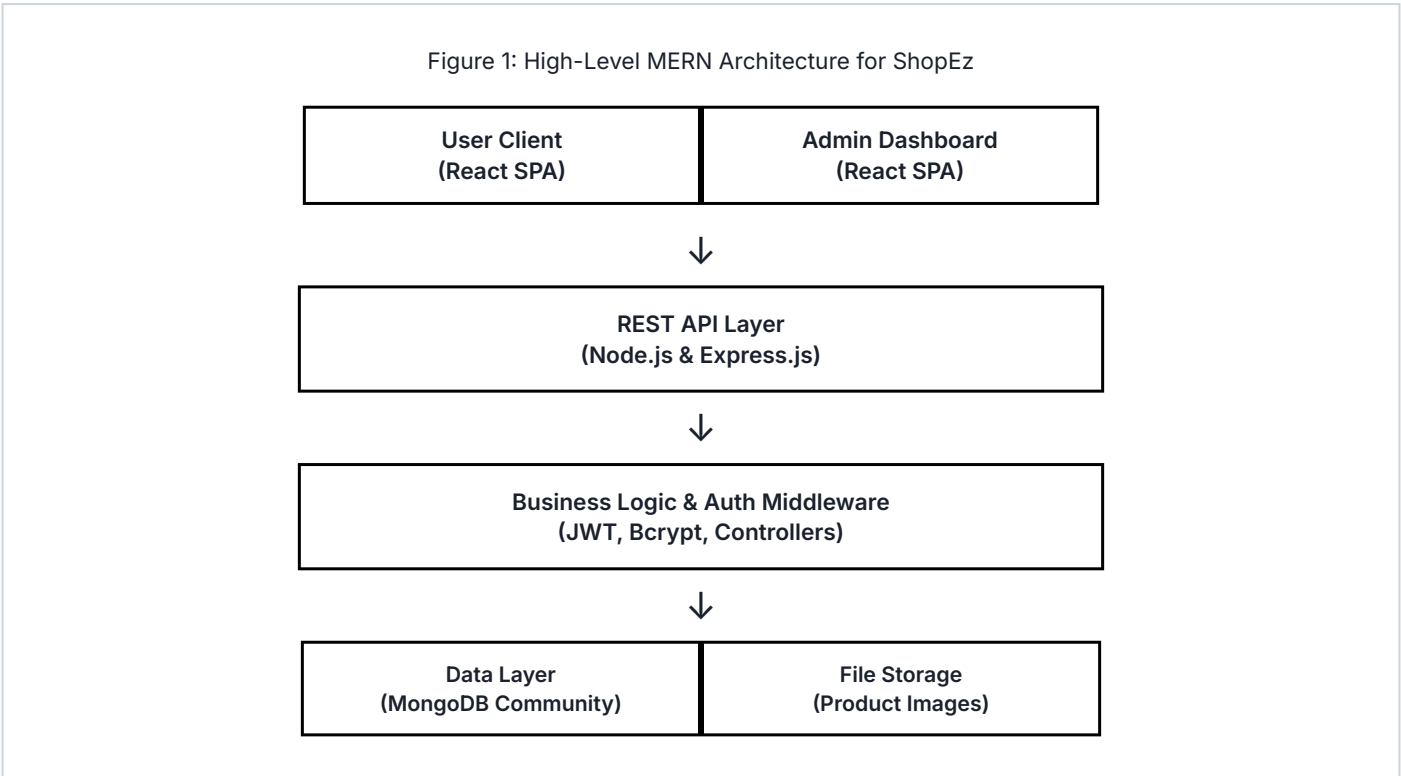
Date	18 June 2026
Team ID	
Project Name	ShopEz (Full Stack MERN-Based E-Commerce)
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture represents a foundational engineering blueprint that aligns technical execution with organizational goals. For the ShopEz platform, the system is structured to provide high scalability for inventory control and robust security for user transactions via the MERN stack.

- Utilize the MERN stack to streamline retail navigation and administrative tasks.
- Implement structured MongoDB schemas for Users, Inventory, and Transactions.
- Guarantee secure communication through Express.js middleware and JWT protocols.
- Enhance user engagement with a fast-loading React-based Single Page Application.

ShopEz - Solution Architecture Diagram:



Component Descriptions:

- Client Layer (React):** Dual-purpose interfaces that offer clean product browsing for shoppers and dedicated administrative tools for catalog management, built with modular UI components.
- API & Server Layer (Node/Express):** Manages secure data traffic, validates user inputs, and handles routing logic to ensure smooth data synchronization via asynchronous backend processes.

Data & Storage Layer (MongoDB): Employs a non-relational document model to manage flexible schemas for customers and sales, integrated with local storage for high-quality product assets.

Logical Data Flow: User actions initiate API requests; the server performs JWT-based session checks, interacts with the MongoDB cluster, and delivers JSON data for dynamic React rendering.